

# Using Snowflake Caching

## Objective

Understand how Snowflake's caching mechanisms improve query performance.

## Query Used

```
SELECT SUM(L_EXTENDEDPRICE)
FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF1.LINEITEM
WHERE L_DISCOUNT > 0.05;
```

## Procedure

### Step 1: Execute Query

Run the query for the first time.

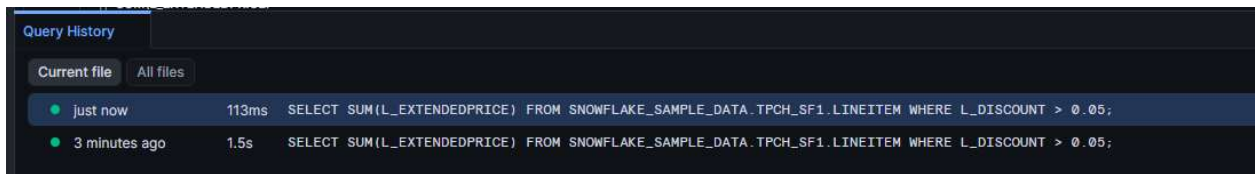
### Step 2: Execute Query Again

Immediately re-run the same query.

### Step 3: Verify Cache Usage

1. Navigate to the **History** tab.
2. Locate both query executions.
3. Examine the **Bytes Scanned** metric.

## Expected Results



| Query History |           |                                                                                                   |
|---------------|-----------|---------------------------------------------------------------------------------------------------|
|               |           |                                                                                                   |
| Current file  | All files |                                                                                                   |
| just now      | 113ms     | SELECT SUM(L_EXTENDEDPRICE) FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF1.LINEITEM WHERE L_DISCOUNT > 0.05; |
| 3 minutes ago | 1.5s      | SELECT SUM(L_EXTENDEDPRICE) FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF1.LINEITEM WHERE L_DISCOUNT > 0.05; |

## Metadata Cache Test

Execute a different query against the same table while accessing different columns.

Example:

```
SELECT AVG(L_QUANTITY)
FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF1.LINEITEM;
```

## Observation

Snowflake leverages metadata cache to improve query planning and execution efficiency.

